

DSCI 471: Final Project Rubric

Final Project Point Distribution

Component	Evaluated Artifact	Point Value
Phase 1	In-Class Presentation	50 Points
Phase 2	Final Report / Jupyter Notebook	300 Points
Total		350 Points

Part A: In-Class Presentation (50 Points)

- **Format:** Brief, free-form presentation (5–7 minutes) to share findings with the class.
- **Grading:** Full credit will be awarded for active participation and an informative presentation of the project's current trajectory (Acceptable as a work-in-progress).

Presentation Score Breakdown

- **Structure & Outline (20 pts):** Follows a logical flow summarizing the core components: Problem Definition → Dataset Exploration → Methodology → Evaluation Metrics.
 - **Clarity & Communication (20 pts):** Delivering a clear, informative presentation that successfully conveys the project's scope, motivations, and current results to peers.
 - **Experiential Reflection (10 pts):** Thoughtfully sharing the teamwork dynamic and practical model training/testing experiences (e.g., debugging challenges, optimization hurdles).
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Part B: Final Report Documentation (300 Points)

Evaluation will heavily weight critical thinking, architecture justification, and proper analytical interpretation over high leaderboard scores or complex raw code.

1. Problem Definition & Clarity (40 Points)

The baseline evaluation of the research definition.

- **Excellent (36–40 pts):** The research problem is perfectly scoped, highly relevant to data science, and backed by strong, clear domain motivations.
- **Proficient (28–35 pts):** The problem is well-defined and practical, though the description of its domain significance or specific research questions could be slightly expanded.
- **Developing (1–27 pts):** The scope is vague, lacks clear real-world motivation, or has minimal connection to data science applications.

2. Data Collection & Exploratory Analytics (50 Points)

The preparation and processing of textual, visual, or numerical sequence data.

- **Excellent (45–50 pts):** Data collection is technically sound. Demonstrates deep exploratory data analysis, appropriate preprocessing or data engineering, and explicit handling of spatial or temporal sequence constraints (e.g., masking, resizing, padding).
- **Proficient (35–44 pts):** The data pipeline is functional and clean. Data traits are adequately explored, but preprocessing steps or feature dimension tracking could be more explicitly documented.
- **Developing (1–34 pts):** Minimal exploratory analysis provided. The dataset lacks the structural scale or documentation necessary to sustain a deep learning pipeline.

3. Intended Methodology & Model Justification (70 Points)

The explicit deep learning structural design using TensorFlow/Keras or other relevant frameworks.

- **Excellent (63–70 pts):** Clear, robust implementation of modern architectures (e.g., CNNs, RNNs, Transformers, or RL). Includes excellent **conceptual justification** for *why* the specific hidden layers, activation functions (e.g., ReLU, GELU), regularization (Dropout, LayerNorm), or model streams were selected for this problem domain.
- **Proficient (49–62 pts):** The model architecture is technically correct and functional. The model selection is logical, but the underlying text lacks deeper commentary explaining the specific structural parameters or layer choices.
- **Developing (1–48 pts):** The methodology lacks technical soundness or relevance to the course. Code blocks appear copied without an understanding of the layers, or the model choices are left completely unjustified.

4. Technical Evaluation & Architecture Comparison (70 Points)

The comparative performance analysis.

- **Excellent (63–70 pts):** Model performance is evaluated using appropriate multi-dimensional tracking metrics (e.g., cross-entropy validation tracking, perplexity when applicable). Includes a structured comparison between different configurations, ablation setups, baseline variants, OR alternative hyperparameter dimensions.
- **Proficient (49–62 pts):** Evaluation metrics are correctly applied and tracked. The model is contrasted across baseline metrics, but the comparison lacks broader architectural depth or comprehensive validation logs.
- **Developing (1–48 pts):** Weak evaluation setup. The project relies on a single metric run without baseline comparisons, or shows an inadequate understanding of proper validation protocols.

5. Interpretation of Results & Critical Thinking (50 Points)

The analytical evaluation of model outcomes.

- **Excellent (45–50 pts):** Provides exceptionally thoughtful interpretation of output tensors and attention maps. Demonstrates clear understanding of whether the system overfitted, underfitted, or hit an architecture-specific ceiling, complete with logical data-driven explanations.
- **Proficient (35–44 pts):** Results are accurately interpreted and discussed. The analysis identifies key performance trends, but stays mostly superficial rather than explaining the underlying model behavior.

- **Developing (1–34 pts):** Missing or flawed interpretation of results. The report provides tables or plots without extracting any meaningful data science insights or analytical conclusions.

6. Technical Documentation & Rigor (20 Points)

The general delivery structure of the project artifact.

- **Excellent (18–20 pts):** The notebook/report is meticulously organized, utilizing clear markdown cell headers and cohesive inline documentation. Code blocks execute flawlessly, and technical terminology is used with high precision.
- **Proficient (14–17 pts):** Clear and readable documentation. Code blocks execute correctly, though formatting consistency or inline explanations could be polished.
- **Developing (1–13 pts):** Poor presentation and messy notebook layout. Contains broken execution graphs, missing asset descriptions, or unreadable blocks of code.